

Summary

Software engineer and Yale PhD researcher building numerical, GPU-accelerated, and distributed applications in Python and C++. Experience spans CUDA/ROCm/HIP, JAX/PyTorch computational graphs, HPC systems, Linux development, and systematic gdb-based debugging.

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Developed graph neural networks and automatic-differentiation models in Python to predict material properties including optical emission spectra.
- Developed Python and C++ software for first-principles and many-body simulation using MPI, OpenMP, PETSc, SLEPc, CUDA, ROCm, and CMake across leadership-class HPC systems.
- Sharded JAX/PyTorch computational-graph workloads across up to 800 Frontier GPUs and TB-scale datasets on national-lab GPU clusters including Frontier, Frontera, and Perlmutter.
- Used gdb for stack, frame, and memory inspection alongside cuda-gdb, nvprof, TotalView, and Linaro DDT for MPI/GPU distributed debugging and performance analysis.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026, demonstrating research output alongside production-style software work.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built parallel applications with CUDA and OpenCL fallback for large sensor-data and waveform processing, diagnostic analytics, and field-data visualization.
- Developed C# and ASP.NET applications for real-time control of company tool testing, data collection, and data processing.
- Developed C# GUI applications for preparing customer inputs and interacting with deployed hardware systems.

FindLight

Photonics Technical Writer

2018–2019

- Wrote technical content translating specialized optics and photonics topics into clear engineering-facing explanations.

Selected Projects

Machine Learning Models for Material Properties: Developed PyTorch Geometric graph neural networks and automatic-differentiation models for molecular-property prediction and optical-properties calculations in material systems.

Agentic Coding and Scientific Automation: Built LangChain-based workflows using OpenAI API models and local Ollama models to read documentation, generate structured inputs, and accelerate coding and research workflows.

Container Orchestration for First-Principles Computing: Developed CUDA-ready and cross-environment containers for first-principles software stacks and published them to Docker Hub for portable scientific computing.

Matrix-Multiply AI Accelerator: Developed a Verilog-based matrix-multiply accelerator project for AI-oriented compute pipelines and hardware-software performance exploration.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University*MS, Electrical Engineering**2023***Cornell University***MEng, Electrical and Computer Engineering**2017–2018***Cornell University***BS, Electrical and Computer Engineering**2013–2017*

Technical Skills

Languages: Python, C, C++, CUDA, HIP, JAX, PyTorch, C#, JavaScript, Bash**Scientific Computing:** MPI, OpenMP, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib**Systems:** Linux, CMake, Docker, Docker Compose, Kubernetes, gcloud**GPU / Debug:** CUDA, ROCm/HIP, cuBLAS, cuTENSOR, cuRAND, rocBLAS, rocFFT, gdb, cuda-gdb, nvprof, TotalView, Linaro DDT**Platforms:** Frontier, Frontera, Perlmutter, Linux, CMake, Docker**ML / Automation:** PyTorch, PyTorch Geometric, scikit-learn, LangChain, OpenAI API, Ollama, graph neural networks, transformers, agentic automation